

TEG

Thermoelectric Generator Selection and Energy Budget

1. Fixed Electrical Load

The selected system components and operating cycle are fixed:

- Activity sensor: Analog Devices ADXL362
- GNSS: Quectel LC76G, active for 15 s every minute
- Radio: Semtech SX1262

Load	Energy per day
LC76G active and power-saving modes	715.5864 J/day
ADXL362 active and standby modes	0.0019 J/day
SX1262 transmit and sleep modes	0.2412 J/day
Total lower-bound load	715.8295 J/day

The corresponding average power is

$$P_{\text{load}} = 715.8295 \text{ J} / 86400 \text{ s} = 8.285 \text{ mW}.$$

This value does not yet include the MCU, memory, power-management quiescent current, or storage losses. The 500 mJ buffer only supports one 495 mJ GNSS event; it does not prove daily energy neutrality.

2. Selected Commercial TEG

The proposed module is the **TECTEG TEG2-126LDT**, which is designed for low-temperature-difference body and sensor harvesting.

Parameter	Value
Module size	40 x 40 x 5.45 mm
Test temperatures	Th = 40 degC, Tc = 30 degC
Module temperature difference	10 K
Matched-load voltage	0.2 V

Parameter	Value
Matched-load current	45 mA
Matched-load power	9 mW

For the theoretical low-temperature range, power is approximated by

$$P_{TEG}(\Delta T) = 9 \text{ mW} \times (\Delta T / 10 \text{ K})^2.$$

3. Proposed TEG Array

The proposed array uses **four TEG2-126LDT modules** in a 2-series, 2-parallel configuration. A 70% average power-conversion efficiency is assumed.

At the selected theoretical design point of $\Delta T_{TEG, rms} = 7 \text{ K}$:

$$\begin{aligned} \text{Power per module} &= 9 \times (7/10)^2 = 4.41 \text{ mW} \\ \text{Array raw power} &= 4 \times 4.41 = 17.64 \text{ mW} \\ \text{Usable power} &= 17.64 \times 0.70 = 12.348 \text{ mW} \end{aligned}$$

Quantity	Result
Usable harvested power	12.348 mW
Fixed load	8.285 mW
Power margin	4.063 mW
Relative margin	49.0%
Usable harvested energy	1066.87 J/day
Fixed load energy	715.83 J/day
Daily energy margin	351.04 J/day

Therefore, the theoretical energy budget closes at a 7 K RMS temperature difference.

4. Minimum Required Temperature Difference

For four modules and 70% conversion efficiency:

$$8.285 \text{ mW} = 4 \times 9 \text{ mW} \times (\Delta T_{req} / 10 \text{ K})^2 \times 0.70$$

$$\Delta T_{\text{req}} = 5.74 \text{ K RMS.}$$

For a 30% power margin, the required value increases to approximately

$$\Delta T_{\text{target}} = 6.54 \text{ K RMS.}$$

The 7 K design point therefore provides slightly more than the desired 30% margin.

5. Temperature-Difference Risk

The Freiburg goat-collar experiment measured only

$$\begin{aligned} \text{mean } \Delta T_{\text{TEG}} &= 2.5 \pm 1.0 \text{ K} \\ \text{observed range} &= \text{approximately } -0.5 \text{ K to } 6 \text{ K.} \end{aligned}$$

At a constant 2.5 K, the proposed four-module array would provide only

$$4 \times 9 \times (2.5/10)^2 \times 0.70 = 1.575 \text{ mW,}$$

which is insufficient. Consequently, feasibility depends on an improved fur-side thermal connector and external heat sink maintaining at least 5.74 K RMS directly across the TEG ceramic surfaces. Body-to-ambient temperature difference must not be substituted for actual TEG temperature difference.

At zero temperature difference, TEG output is zero. Energy storage must bridge these periods:

Zero-output duration	Required stored load energy
12 h	357.9 J = 0.099 Wh
24 h	715.8 J = 0.199 Wh
7 days	1.39 Wh

6. Conclusion

The selected theoretical solution is **4 x TECTEG TEG2-126LDT**. It supplies approximately 12.35 mW usable power at a 7 K RMS module temperature difference, compared with the

8.285 mW fixed lower-bound load, giving a 49% power margin.

The electrical energy budget is therefore conditionally feasible. The decisive design condition is maintaining at least 5.74 K RMS across the TEGs; this thermal condition, rather than the catalog power rating, is the main project risk.

Sources

- TECTEG, TEG2-126LDT low-temperature-difference module: <https://tecteg.com/low-dt-thermoelectric-harvesting-teg-power-module/>
- Baeumker et al., "A Fully Featured Thermal Energy Harvesting Tracker for Wildlife," Energies 2021: <https://doi.org/10.3390/en14196363>